

A Statistical Toolkit for Linguistic Typology: Semantic maps

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Plan (tentative)

- Session 1: testing cross-linguistic generalizations
- **Session 2: creating semantic maps**
- Session 3:
 - comparing constraints
 - discussing your own ideas

Course materials

See the slides, data and code at
<https://github.com/levshina/Frejus2026>

(Session 2)

OR in the shared directory (see email)

Outline

- Connectivity maps
 - Main principles
 - A hands-on case study with R: colexifications of FOOT
- Proximity maps
 - Main principles
 - A hands-on case study with R: European analytic causatives

Overviews

- Georgakopoulos, T., & Polis, S. (2018). The semantic map model: State of the art and future avenues for linguistic research. *Language and Linguistics Compass* 12: e12270. <https://doi.org/10.1111/lnc3.12270>
- Haspelmath, M. (2003). The geometry of grammatical meaning: Semantic maps and cross-linguistic comparison. In M. Tomasello (Ed.), *The new psychology of language*, vol. 2 (pp. 211-242). Lawrence Erlbaum.
- Levshina, N. 2017. SÉMANTICKÁ MAPA (Semantic maps). In P. Karlík, M. Nekula, J. Pleskalová (eds.) *Nový encyklopedický slovník češtiny* (New Encyclopedic Dictionary of Czech). Nakladatelství Lidové noviny. English version: https://levshina.github.io/docs/Levshina2017_SemanticMaps.pdf
- van der Auwera, J. (2013). Semantic maps, for synchronic and diachronic typology. In A. Giacalone Ramat, C. Mauri & P. Molinelli (Eds.), *Synchrony and diachrony: a dynamic interface* (pp. 153-176). Benjamins.

Semantics and space

- Semantic **fields**
- **Near**-synonymy
- semantic **contiguity**
- Vector **space** models
- **cosine** measures of similarity
- **broadening** and **narrowing**
- ...

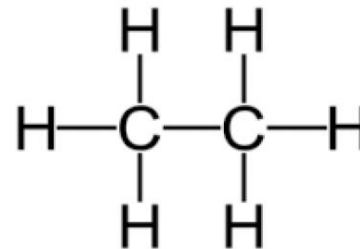
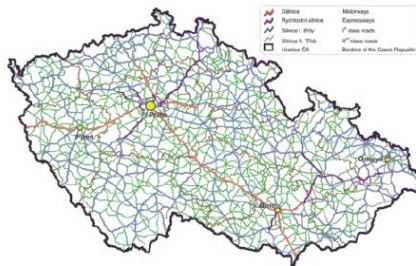
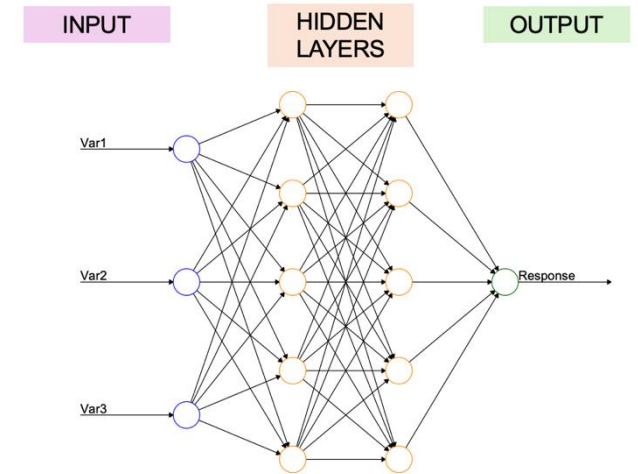
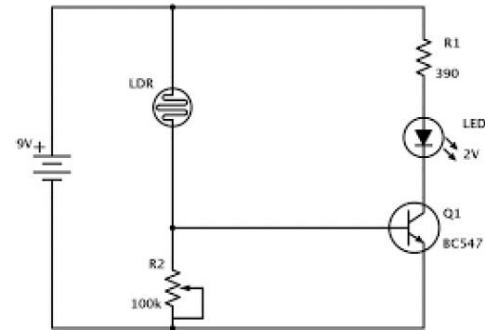
Semantic maps

- Represent a convenient tool for comparison of semantic and pragmatic functions across languages
- Based on different kinds of data:
 - grammars and typological databases,
 - dictionaries
 - parallel corpora
 - experimental data

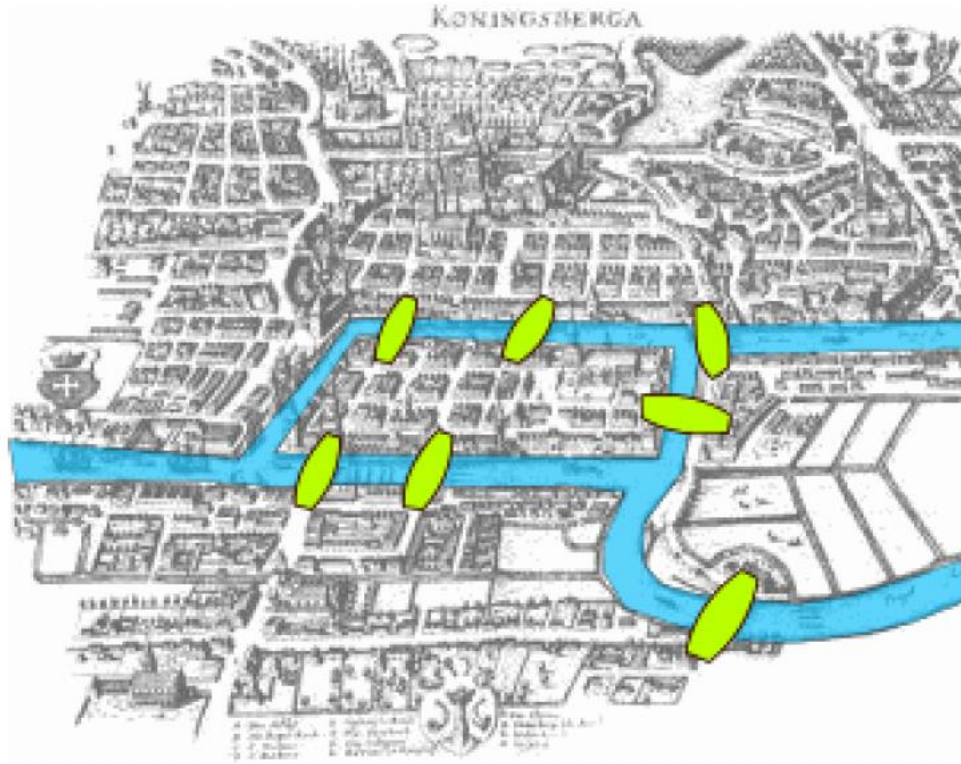
Main types of semantic maps

- **Connectivity maps** are graphs, where meanings are nodes (vertices) connected by links (edges).
 - Undirected vs. directed (e.g. diachrony)
 - Weighted vs. unweighted
- **Probabilistic maps**, with distances between objects (e.g. examples from a parallel corpus or stimuli in an experiment)

Graphs are everywhere!



Seven bridges of Königsberg

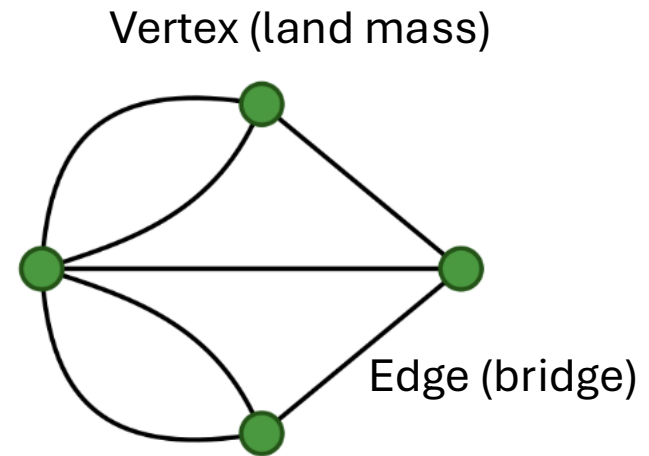


Devise a walk through the city that would cross each of the bridges once, and only once.

Graph theory



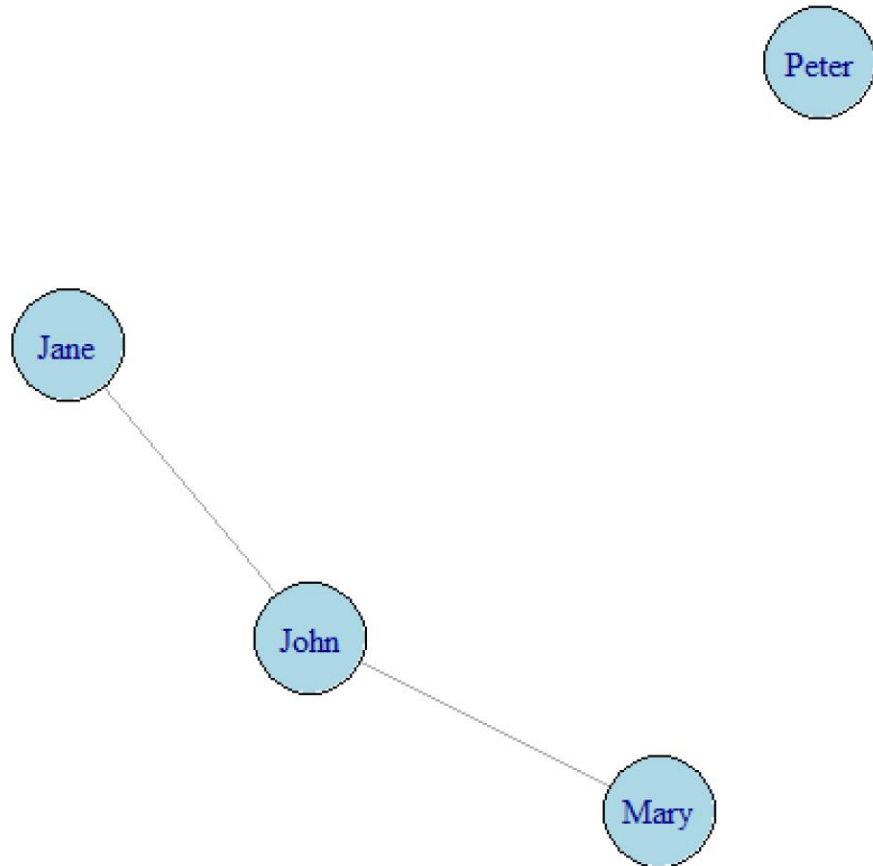
Leonhard Euler (1707 -1783)



Main types of graphs

- Undirected
 - Directed
-
- Unweighted
 - Weighted

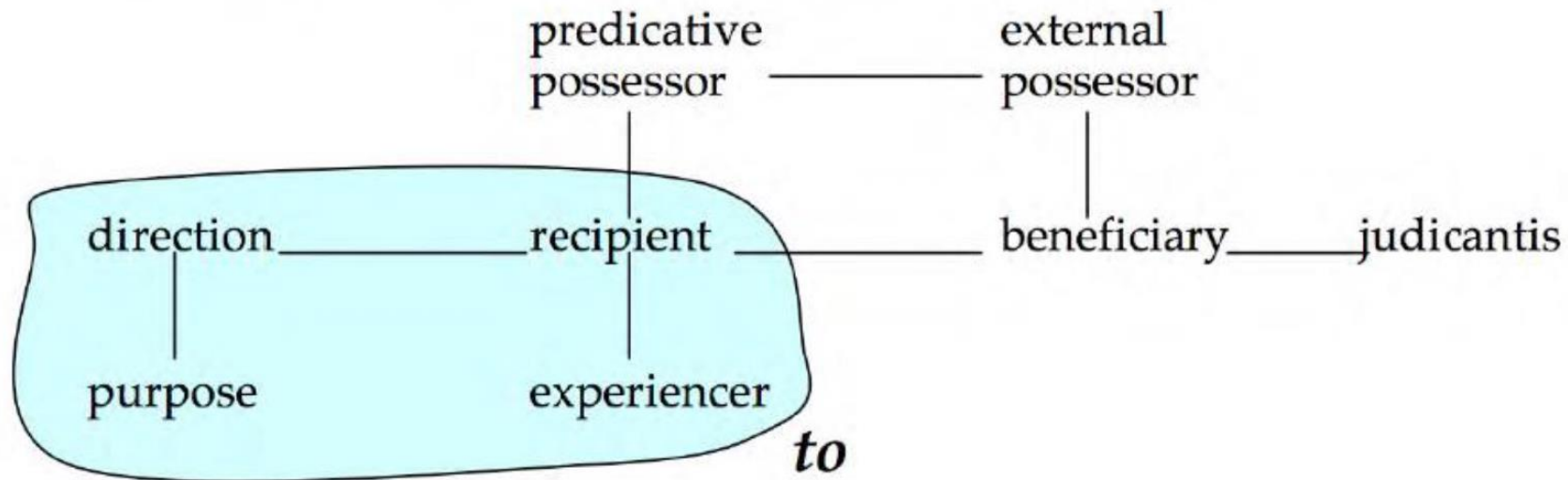
Undirected graphs



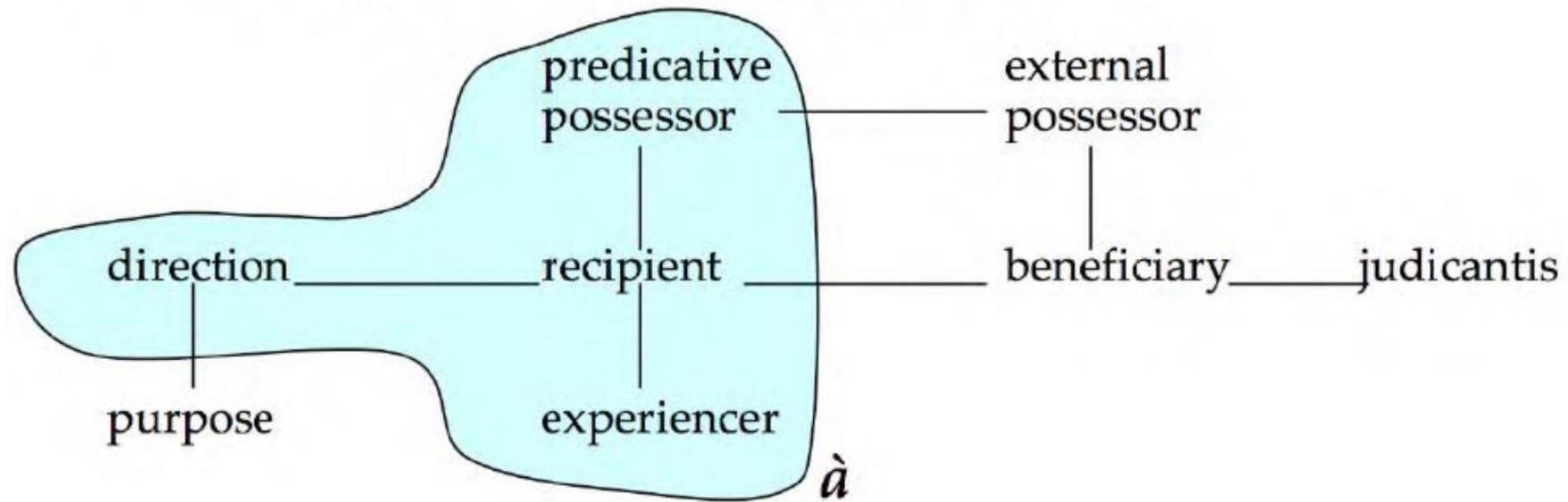
- Peter is single
- John and Mary are married.
- Jane is John's ex-wife.

Haspelmath (2003): datives

the English **Dative** preposition *to*



the French **Dative** preposition *à*



Main principles of connectivity maps

- Nodes (vertices): A function is put on a map when there's at least one pair of languages which differ wrt. this function (Haspelmath 2003)
- Links (edges): the principle of connectivity (adjacency/contiguity): if a construction has more than one function, they should be connected (see van der Auwera 2013).

Identification of nodes: TREE/WOOD/FOREST

Danish

Træ

Skov

from Georgakopoulos & Polis 2018, based on Hjelmslev 1961

Identification of nodes: TREE/WOOD/FOREST

Danish	French
<i>Træ</i>	<i>arbre</i>
	<i>bois</i>
<i>Skov</i>	<i>Forêt</i>

from Georgakopoulos & Polis 2018, based on Hjelmslev 1961

Identification of nodes: TREE/WOOD/FOREST

Danish	French	German
<i>Træ</i>	<i>arbre</i>	<i>Baum</i>
	<i>bois</i>	<i>Holz</i>
<i>Skov</i>		<i>Wald</i>
	<i>Forêt</i>	

from Georgakopoulos & Polis 2018, based on Hjeltmslev 1961

Identification of nodes: TREE/WOOD/FOREST

		Lexical items			
		Danish	French	German	Spanish
ANALYTICAL PRIMITIVES	TREE	Træ	arbre	Baum	árbol
	WOOD (mat.)		bois	Holz	madera
	FIREWOOD				leña
	FOREST (small)	Skov	Forêt	Wald	bosque
	FOREST (large)				selva

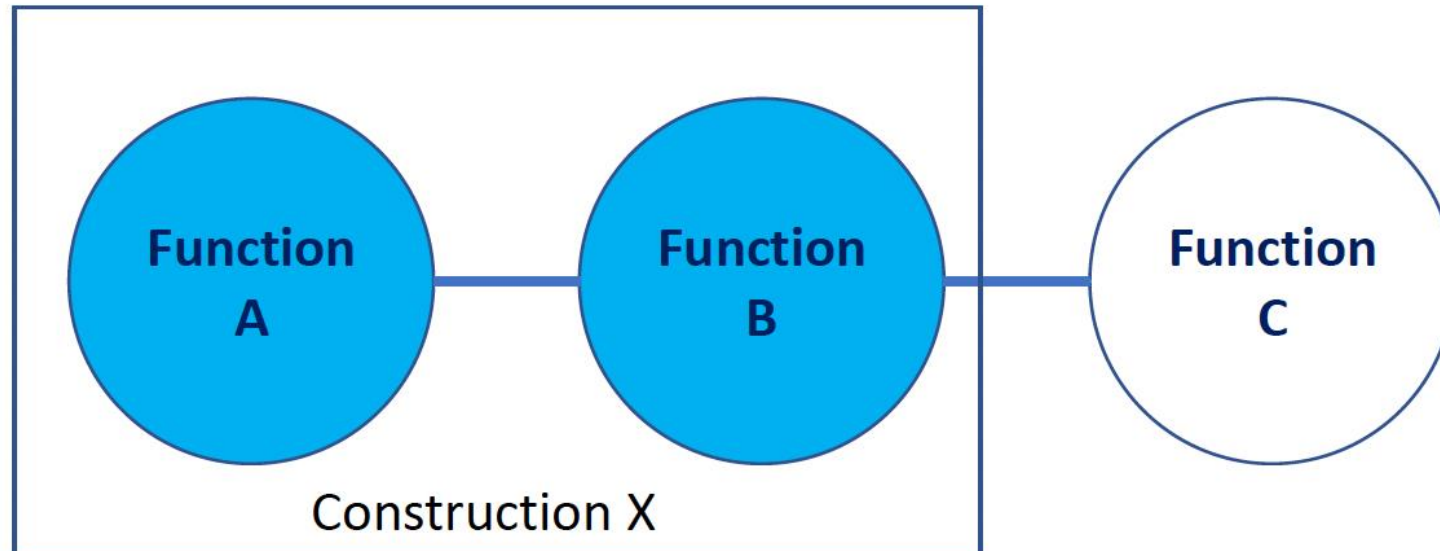
from Georgakopoulos & Polis 2018, based on Hjelmslev 1961

Identification of nodes: TREE/WOOD/FOREST

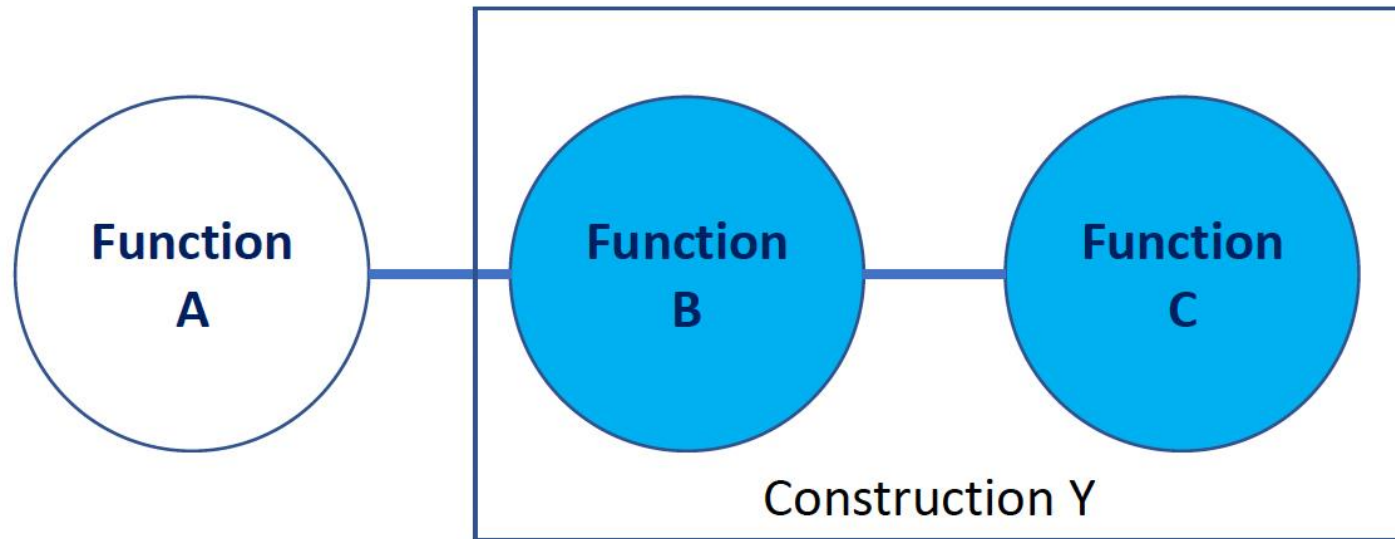
		Lexical items			
		Danish	French	German	Spanish
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	WOOD (mat.)		bois	Holz	madera
	FIREWOOD				leña
	FOREST (small)	Skov	Forêt	Wald	bosque
	FOREST (large)				selva

Do we need more nodes?

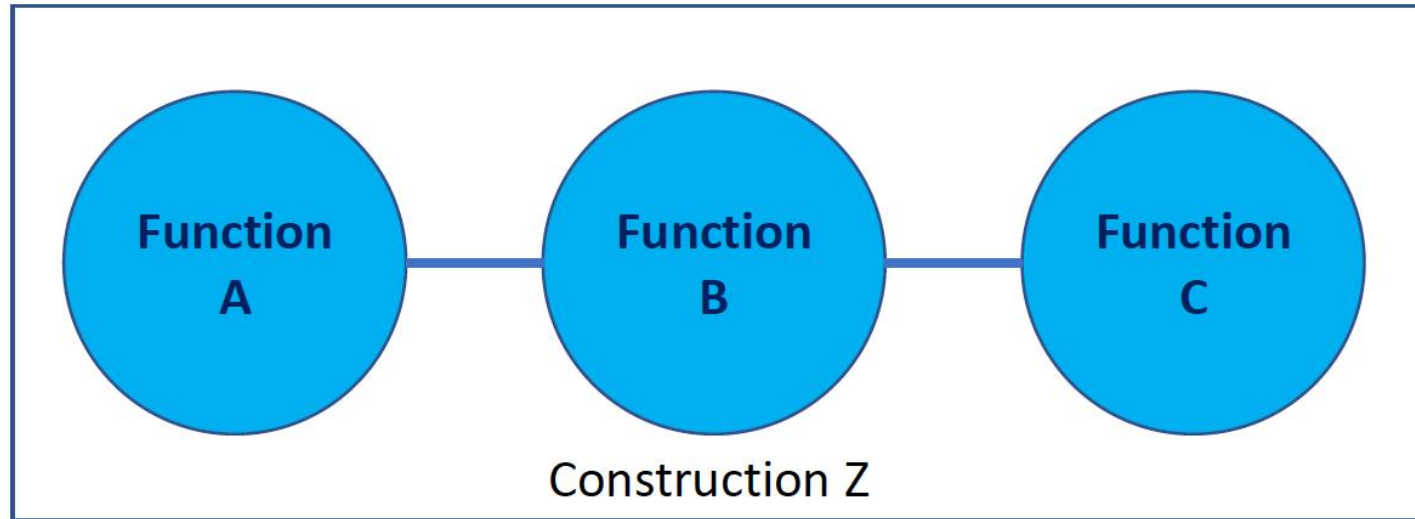
Connecting the nodes



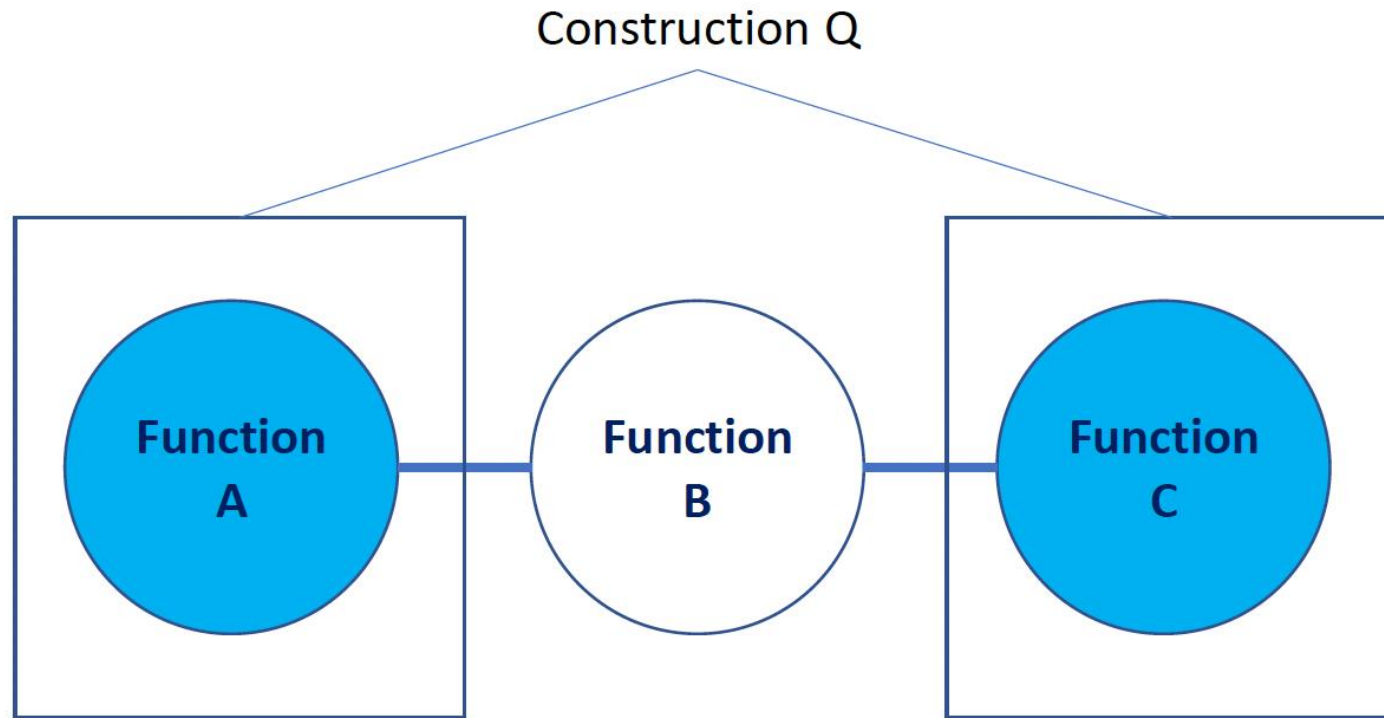
Connecting the nodes



Connecting the nodes

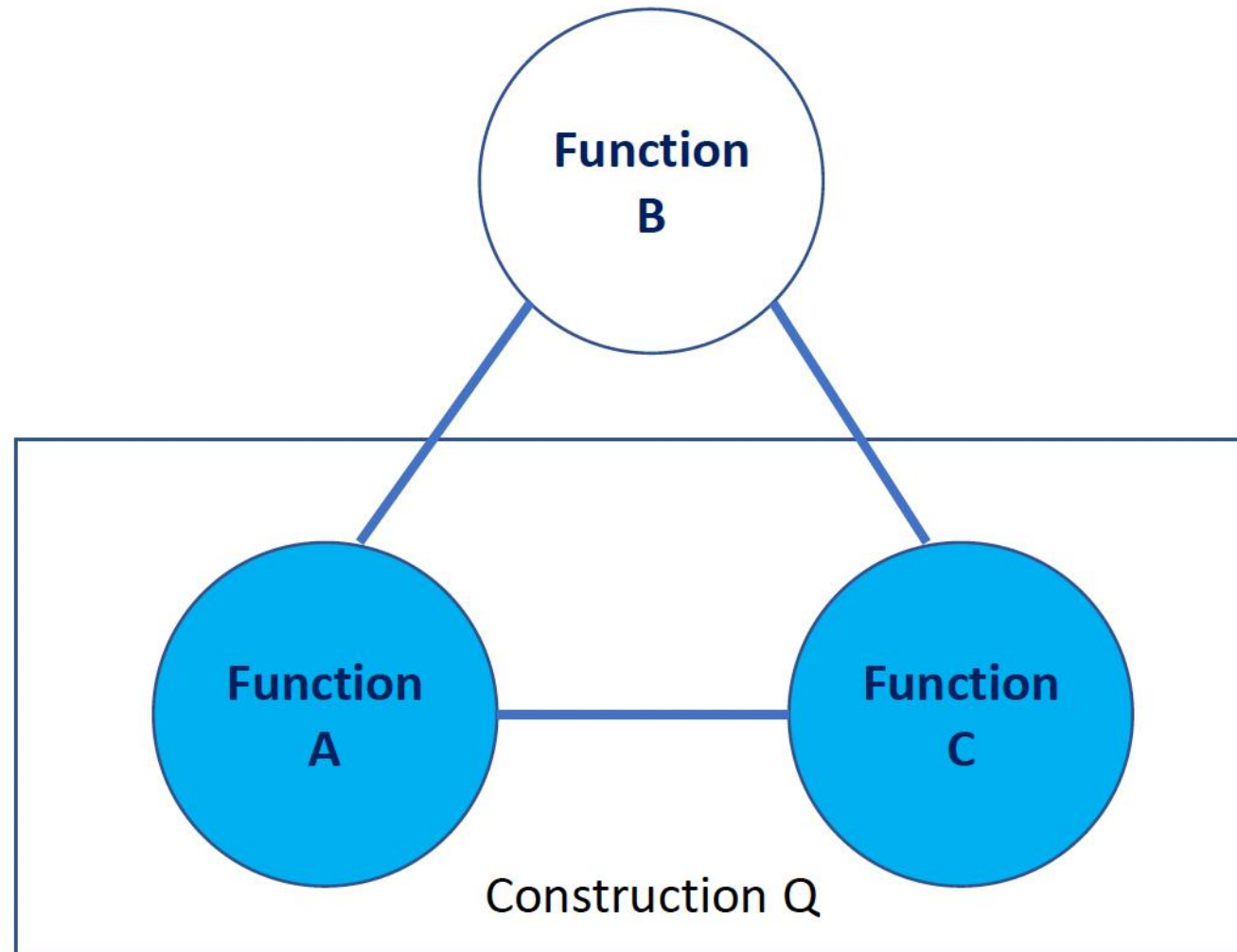


Connecting the nodes



Wrong:
the connectivity principle is not observed!

A fix



Parsimony

- "We will consider a map to be high-quality if it is relatively parsimonious – i.e., if it accommodates the data using few edges..." (Regier et al. 2013)
- A theoretical model: hypotheses about paths of semantic change

An activity

- What would a connectivity map of TREE/WOOD/FOREST look like, based on Hjelmslev's data?
- Test if it works on other languages you know.

		Lexical items			
		Danish	French	German	Spanish
ANALYTICAL PRIMITIVES	TREE	<i>Træ</i>	<i>arbre</i>	<i>Baum</i>	<i>árbol</i>
	WOOD (mat.)		<i>bois</i>	<i>Holz</i>	<i>madera</i>
	FIREWOOD				<i>leña</i>
	FOREST (small)	<i>Skov</i>	<i>Forêt</i>	<i>Wald</i>	<i>bosque</i>
	FOREST (large)				<i>selva</i>

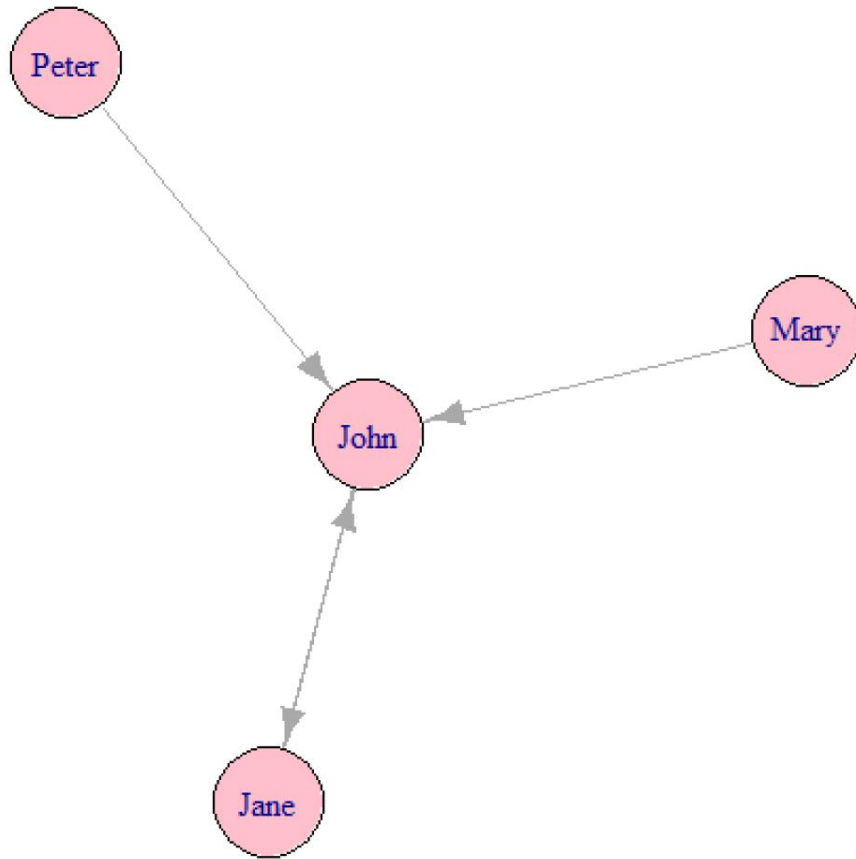
Automatic methods

- It has been shown that the inference problem of creating such a map is computationally intractable (Croft & Poole 2008; Regier et al. 2013).
- However, one can use an approximation, such as the efficient algorithm for the social network inference problem developed by Angluin et al. (2010).
 - Regier et al. (2013): map of spatial semantics
 - Levshina (2022) <https://doi.org/10.1515/zfs-2021-2043>: a pilot map of causative constructions

Controversy: cognitive status

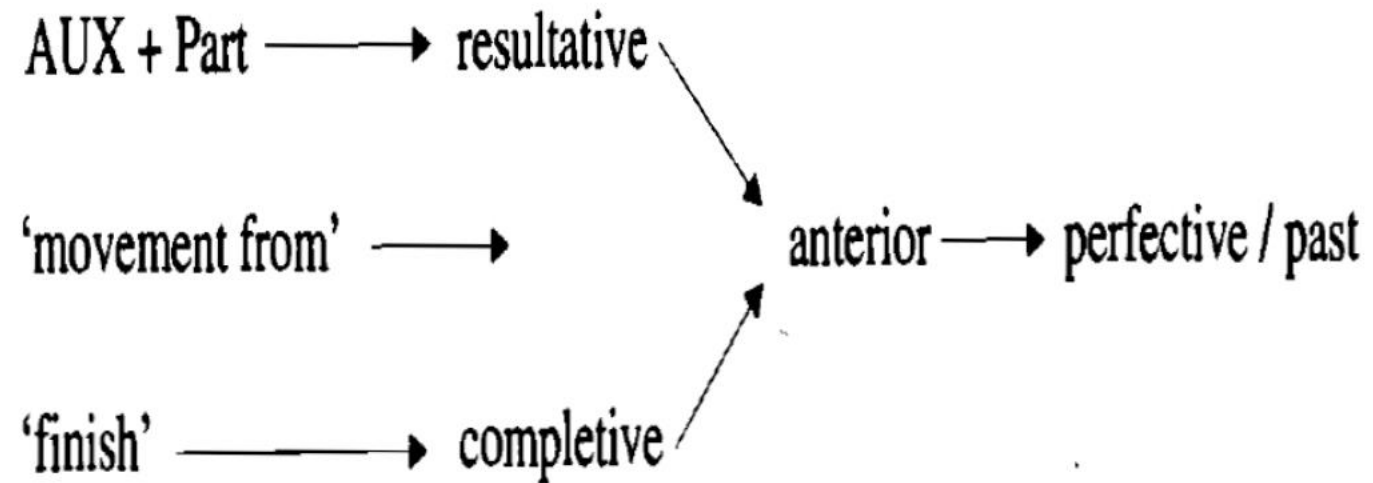
- A semantic map reflects universal organization of conceptual knowledge across languages, a 'geography of the human mind' (Croft 2001)
- Scepticism (Cristofaro 2010)
- How to test?

Directed graphs



- Mary loves John.
- John loves Jane.
- Jane loves John.
- And Peter fancies John, too.

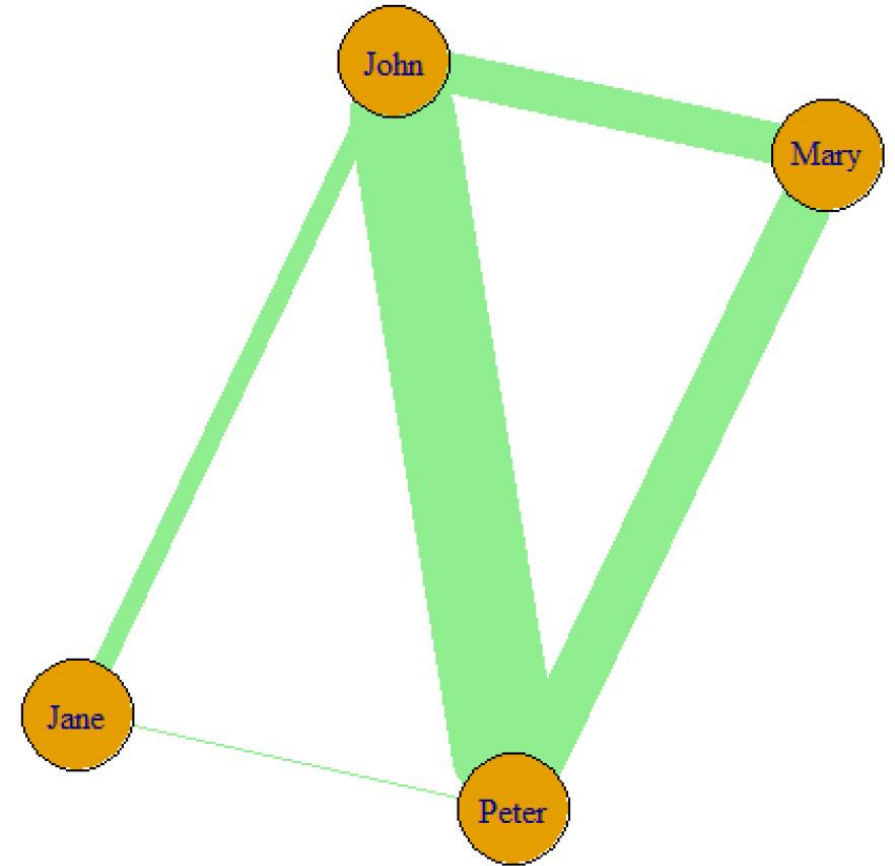
Tense and Aspect Grams



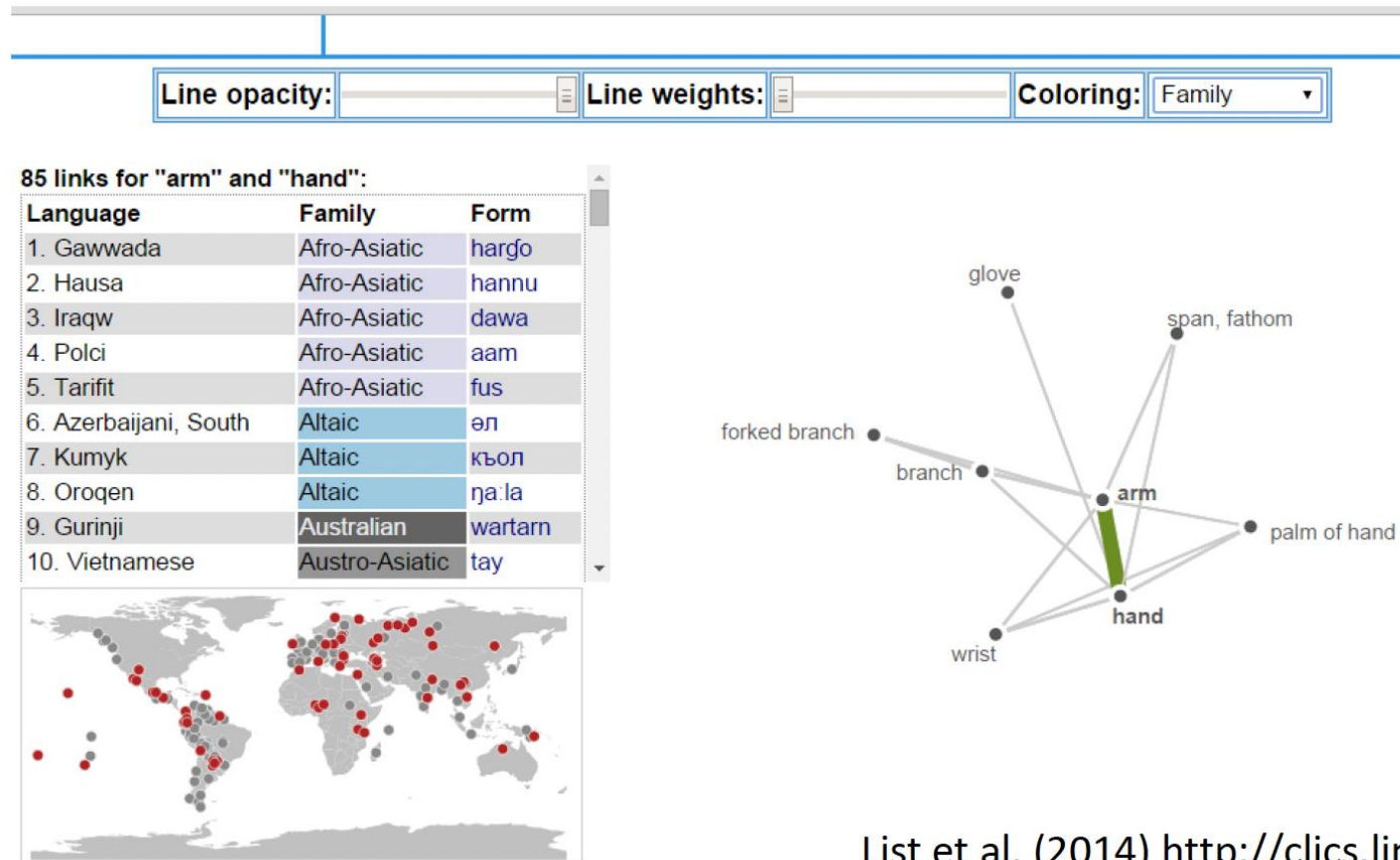
Bybee et al. 1994

Weighted graphs

- Mary and John have eaten 20 pizzas.
- John and Jane have eaten 10 pizzas.
- Peter and John have eaten 50 pizzas.
- Mary and Peter have eaten 25 pizzas.
- Jane and Mary have eaten 0 pizzas.
- Jane and Peter have eaten 1 pizza.



Colexification patterns of HAND



List et al. (2014) <http://clics.lingpy.org/>

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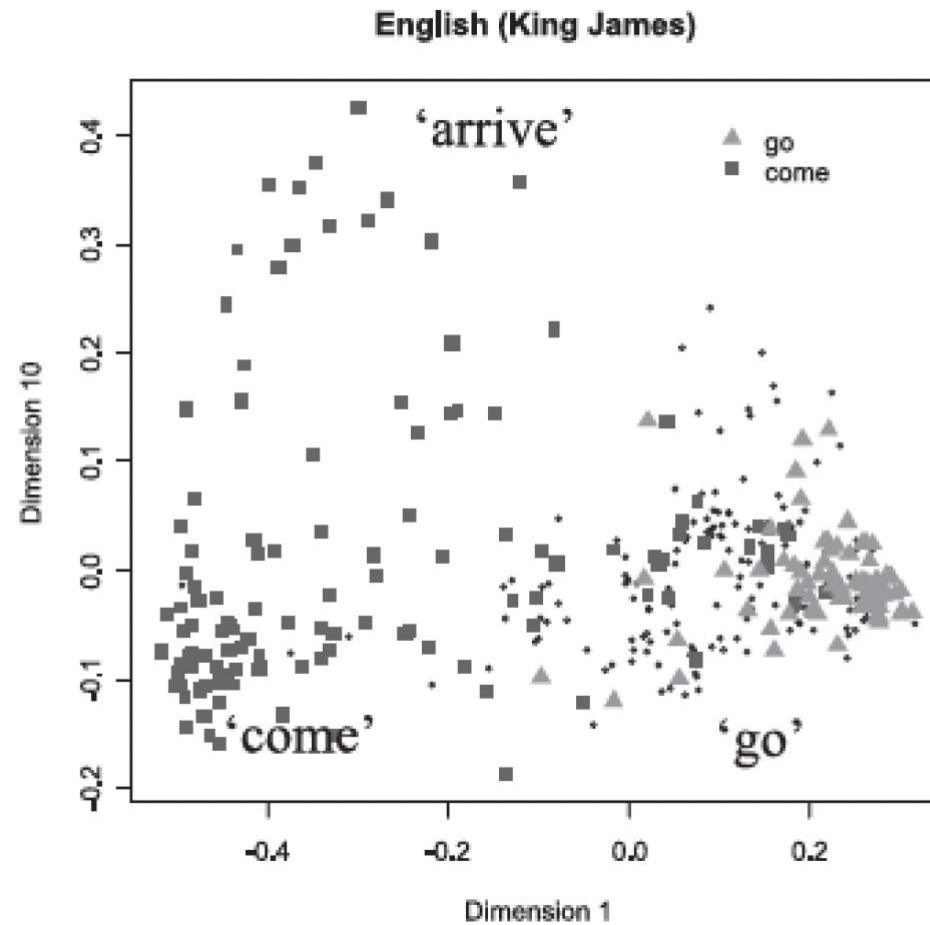
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Proximity maps

- Probabilistic semantic maps are often based on parallel corpora.
- The main unit is a specific context in a parallel corpus (e.g. sentence from the New Testament translated into many languages), not a concept or semantic function.
- The relationships between contexts are represented as distances in MDS, not links.
- The distances between contexts are determined by the similarity between the translations (forms) in the documents.

Wälchli & Cysouw (2012): motion verbs



The causative continuum

Lexical <> Morphological <> Analytic (Periphrastic)

e.g. *kill, break*

e.g. Turkish *öldür*- “kill”
from *öl*- “die”

e.g. *cause* X to die,
make X disappear

ParTy corpus of film subtitles

- Translations are aligned with the English version at the level of sentences
 - Jörg Tiedemann's software subalign with manual correction
- Available at <https://github.com/levshina/ParTy-1.0>
- ParTree: Parsed with UD, expanded

Ebert, C., Levshina, N., & Widmer, P. (2023). ParTree - Parallel Treebanks: A multilingual corpus of movie subtitles. (Version 1.0.0) [Data set]. LaRS - Language Repository of Switzerland. <https://doi.org/10.48656/zjzp-gj69>

Dataset

- This case study uses data from 10 languages
 - English, Finnish, French, Hebrew, Indonesian, Mandarin Chinese, Russian, Thai, Turkish, Vietnamese
- I selected 182 causative events using a very broad definition of causation (including permissive, non-implicative, accidental, assistive causation, etc.)

Films



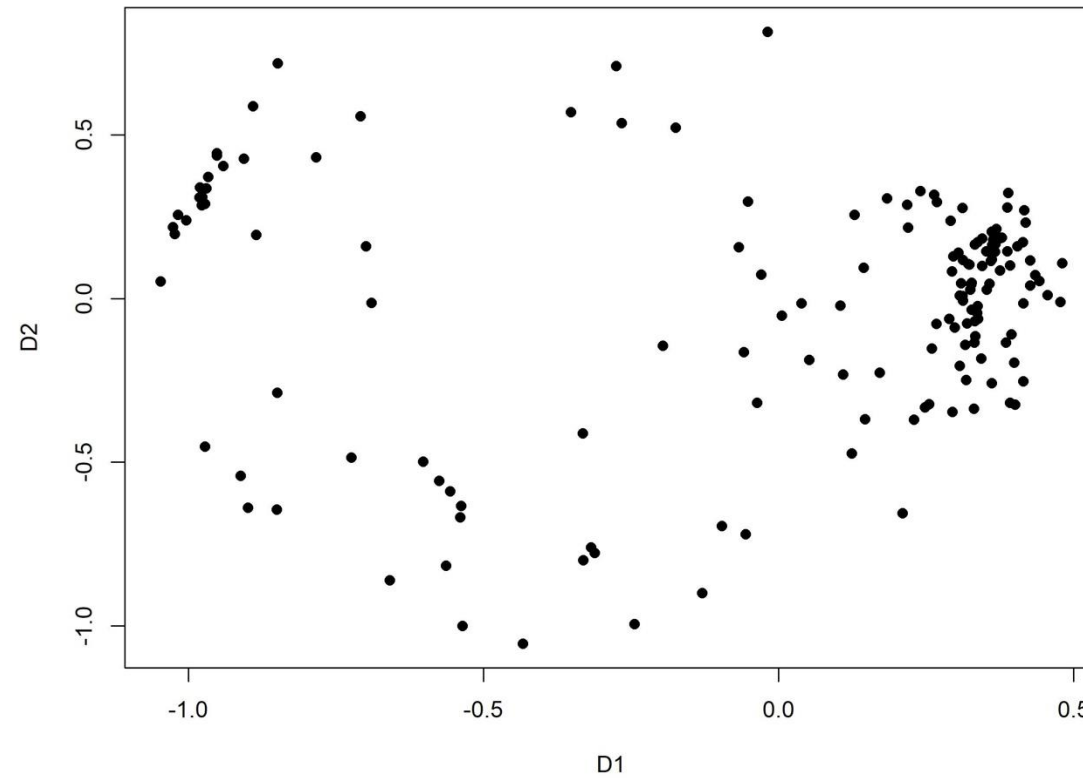
Levshina 2022

Context x language matrix (fragment)

Film	Context	RUS	FIN	FRA	THA	VIE
Avatar	They can fix the spine, if you got the money.	Lex	Lex	Lex	Lex	Lex
Avatar	Let's go special case, do not make me wait for you.	zastavljat'_V	Morph	faire_V	hai_Pred	để_Pred
Avatar	...that will stop your heart in one minute.	Lex	Morph	NA	Lex	NA
Avatar	As head of security, it is my job to keep you alive.	NA	pitää_V	NA	NA	NA
Avatar	It's nothing like an old school safety brief to put your mind at ease.	Lex	Lex	mettre_PP	Lex	NA

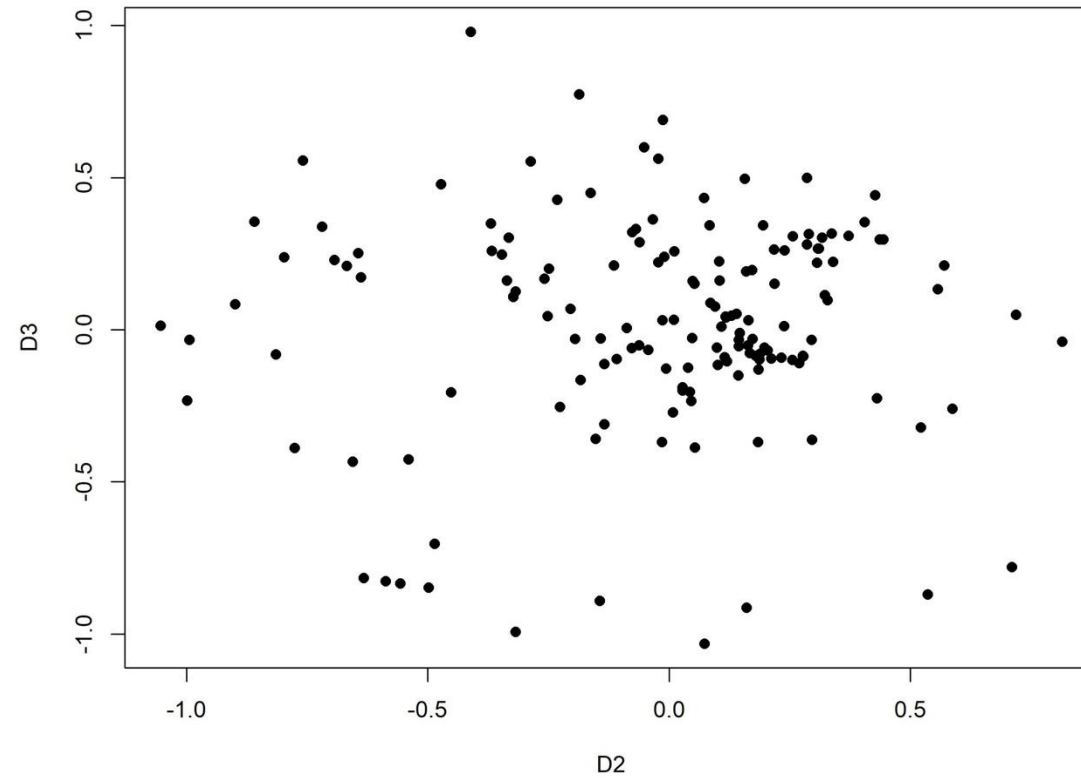
Levshina 2022

MDS: Dimensions 1 & 2



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MDS: Dimensions 2 & 3



Levshina 2022

Interpretation of dimensions

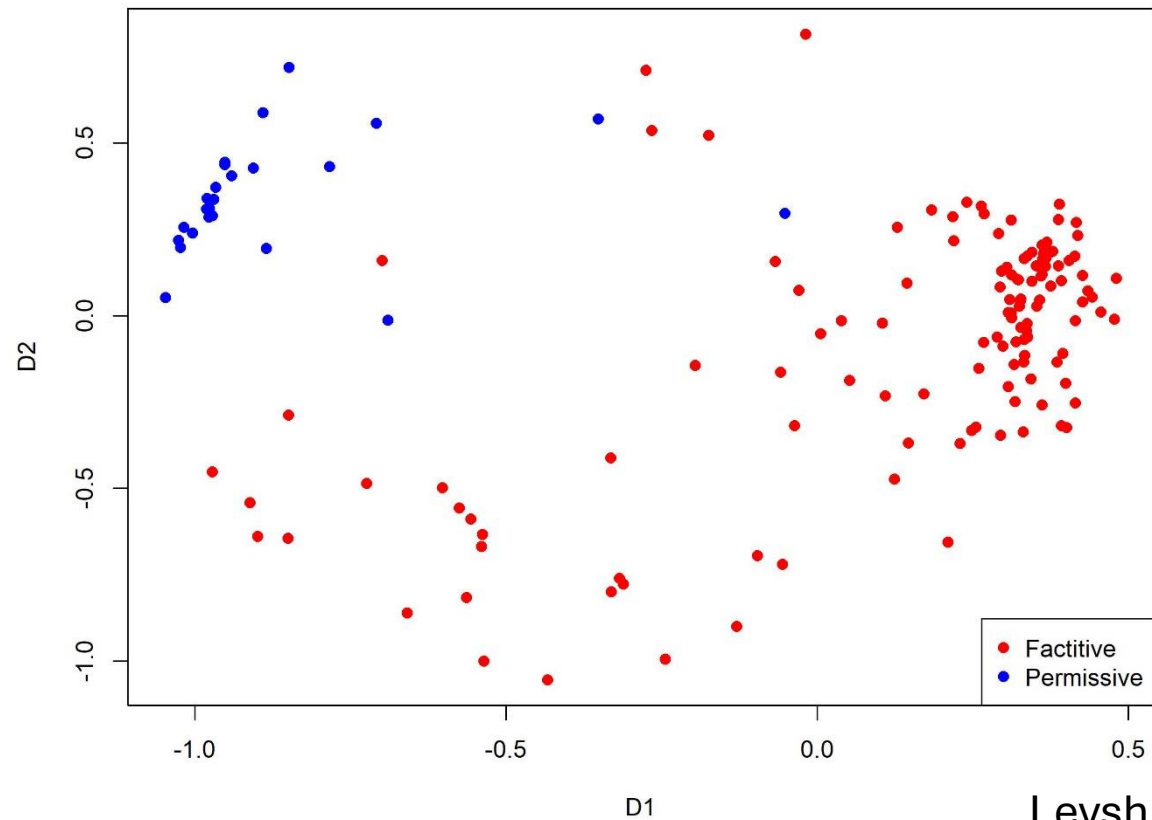
Semantic distinction	Dimension 1	Dimension 2	Dimension 3
Intentional or Not	-0.004	0.071	0.032
Forceful or Not	0.011	0.004	-0.005
Comitative or Not	0.003	-0.004	0.001
Direct or Not	0.191	-0.004	0.000
Causee Control or No Control	0.203	-0.001	0.015
Factitive or Permissive	0.520	0.158	0.045
Caused Event or State	0.290	-0.003	0.012
Implicative or Not	0.021	-0.006	0.037
Assistive or Not	0.019	0.022	0.147
Curative or Not	0.005	-0.005	-0.005
Directive or Not	0.021	-0.006	0.038

The numbers are the R^2 of linear regression model with the coordinates of the points as the response and a particular semantic distinction.

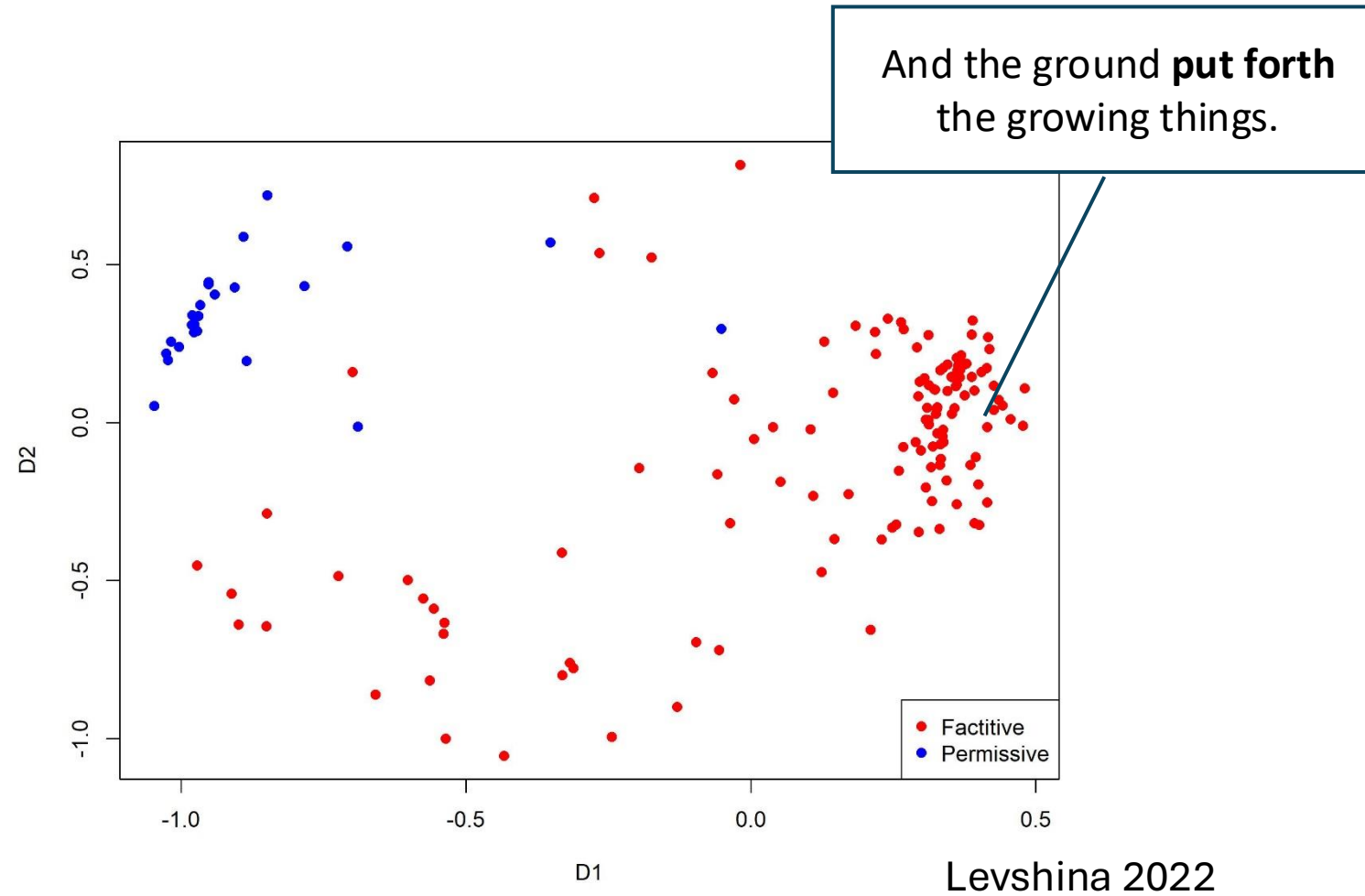
Interpretation of dimensions

- The R^2 show that the 1st and 2nd dimensions both are the most strongly associated with the contrast between factitive causation (i.e. making X do Y) and permissive causation (i.e. letting X do Y).

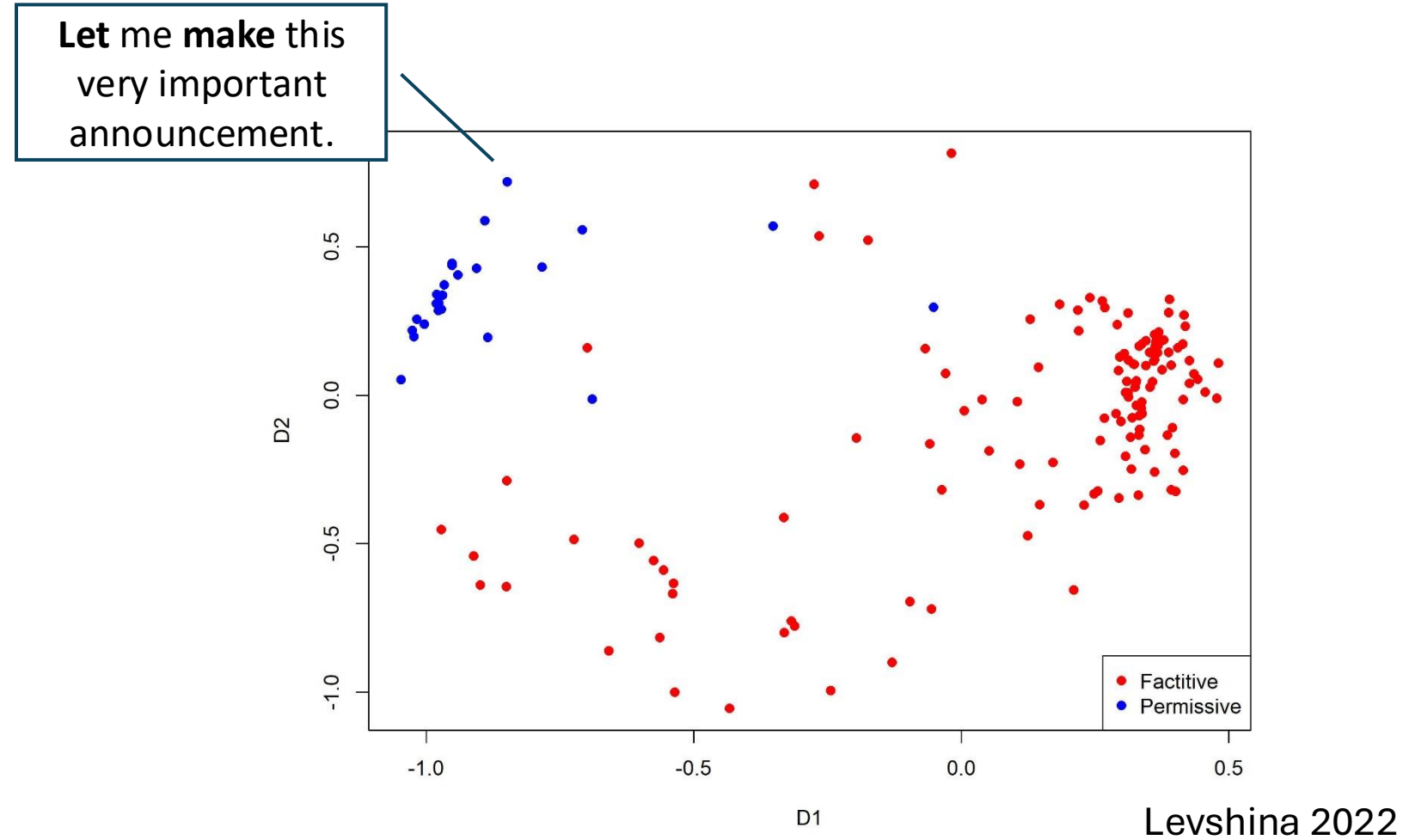
Dimensions 1 & 2: factitive (making) vs. permissive (letting)



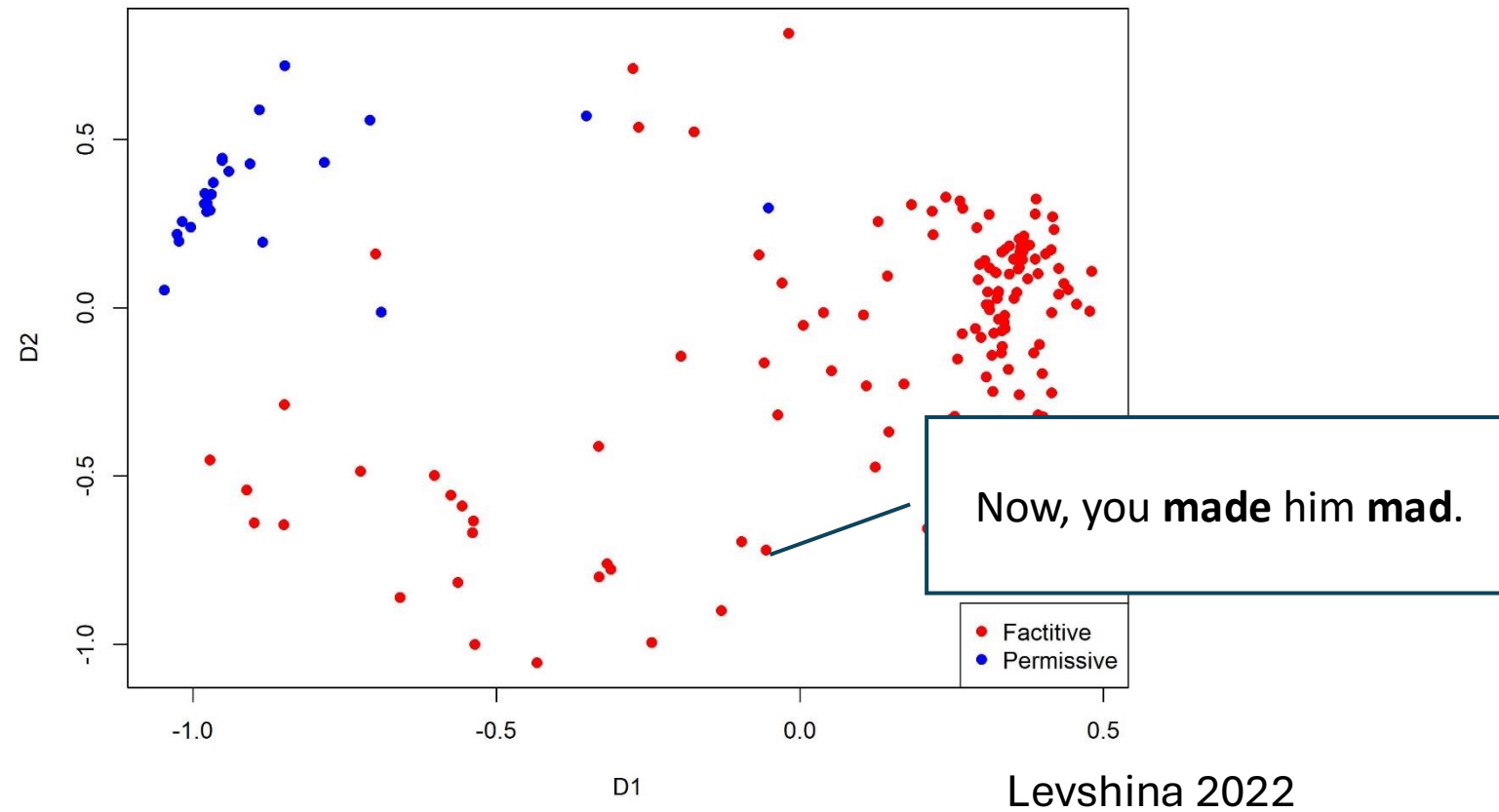
Dimensions 1 & 2: contexts



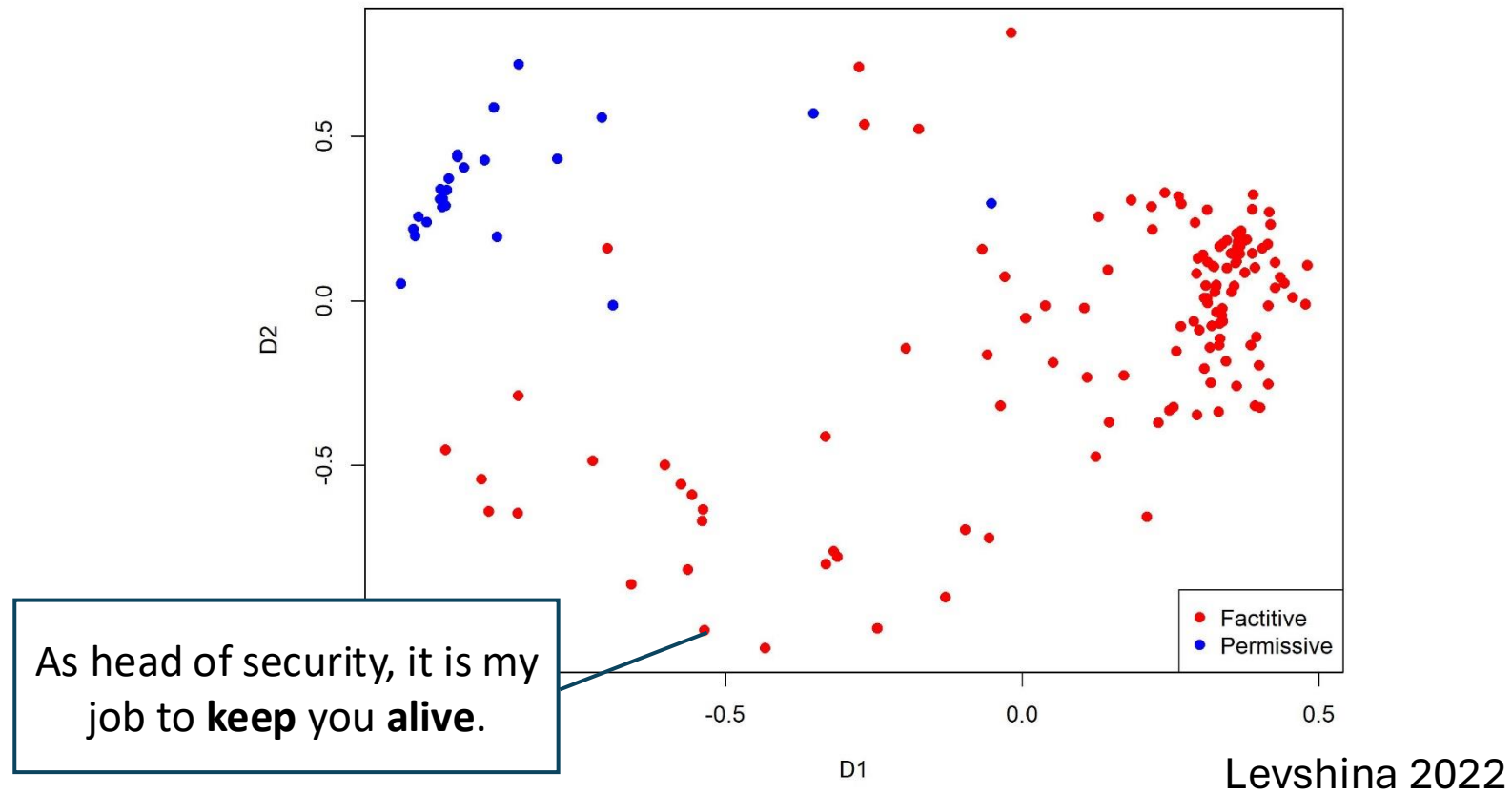
Dimensions 1 & 2: contexts



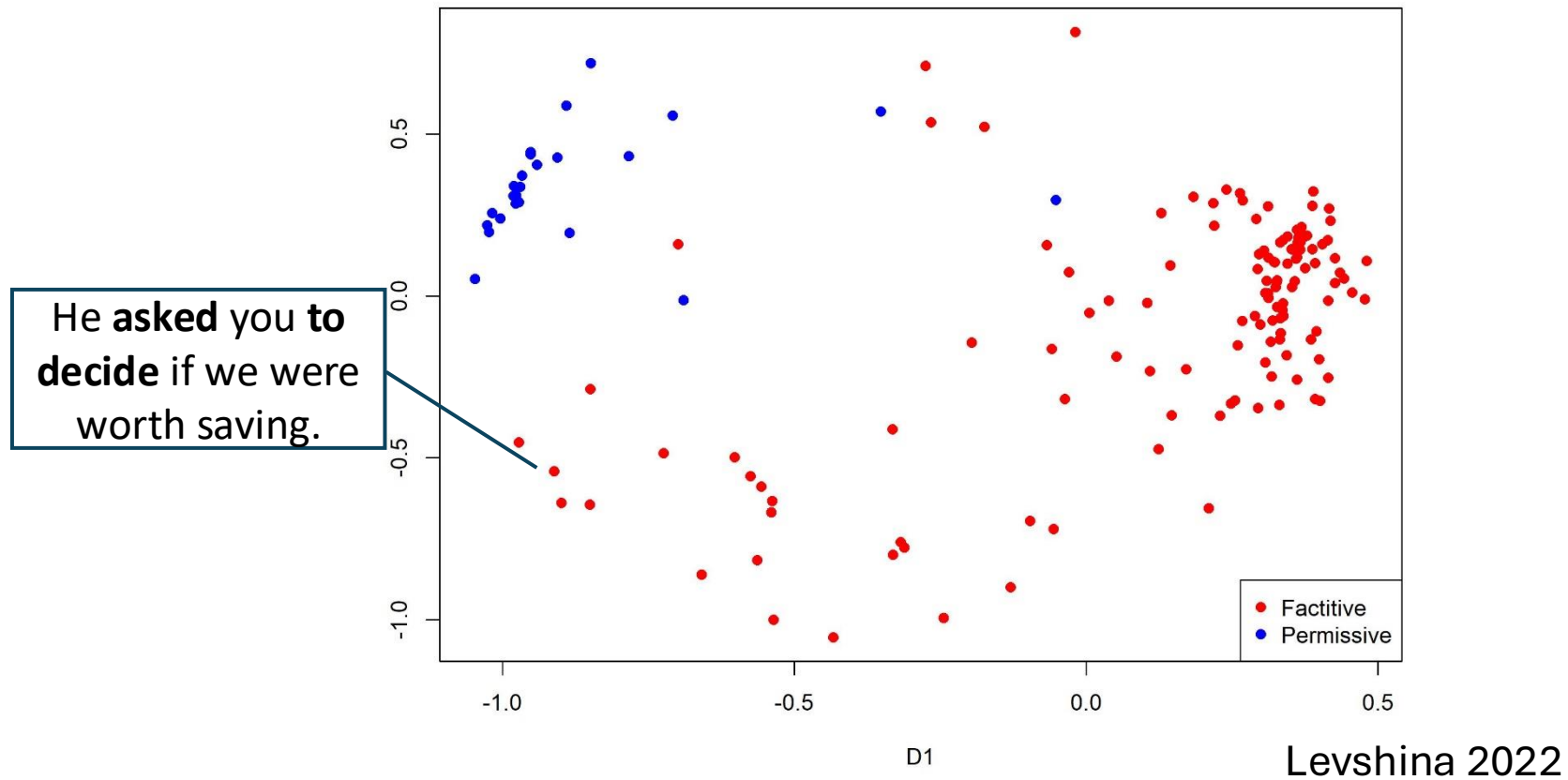
Dimensions 1 & 2: contexts



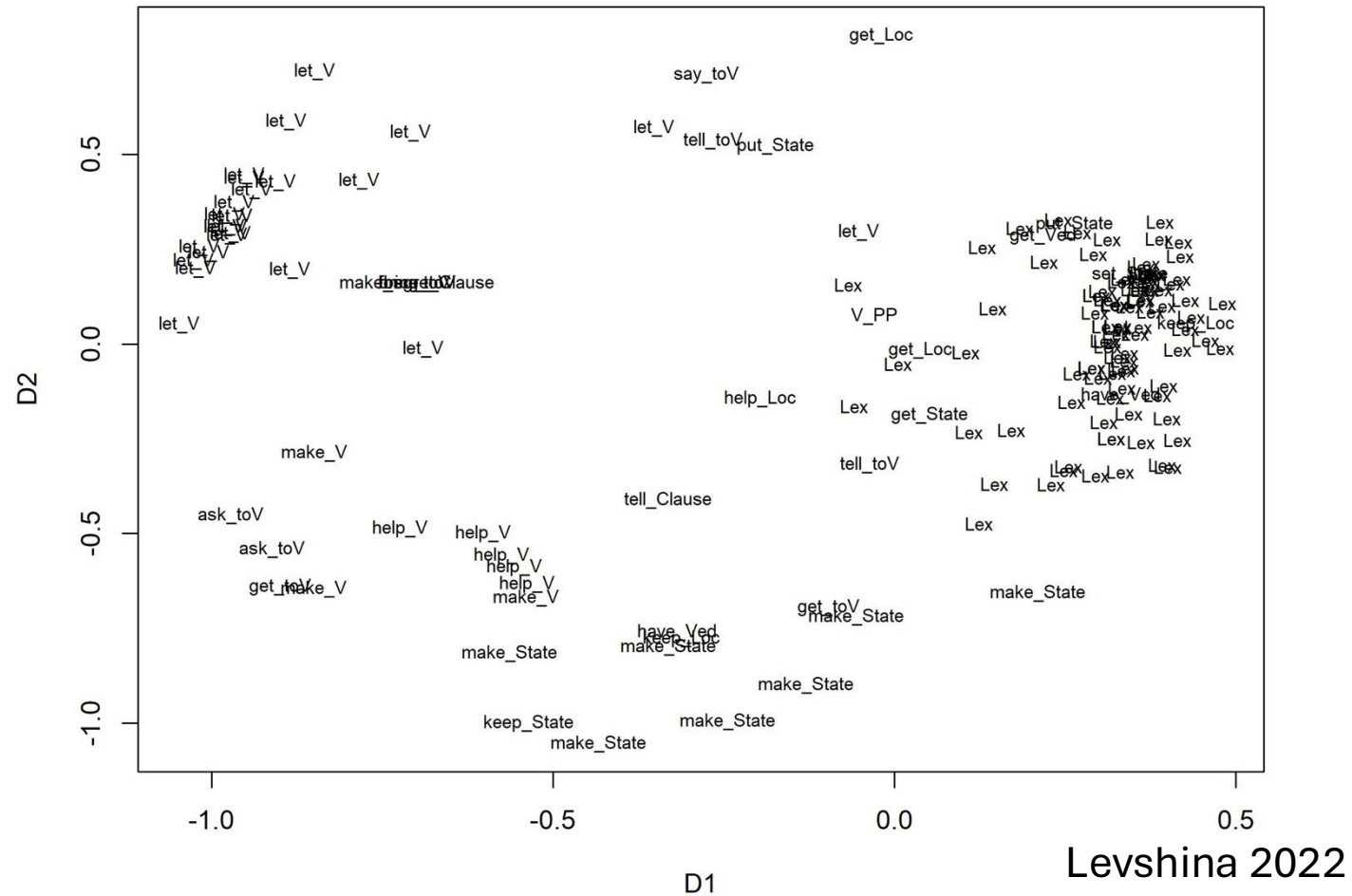
Dimensions 1 & 2: contexts



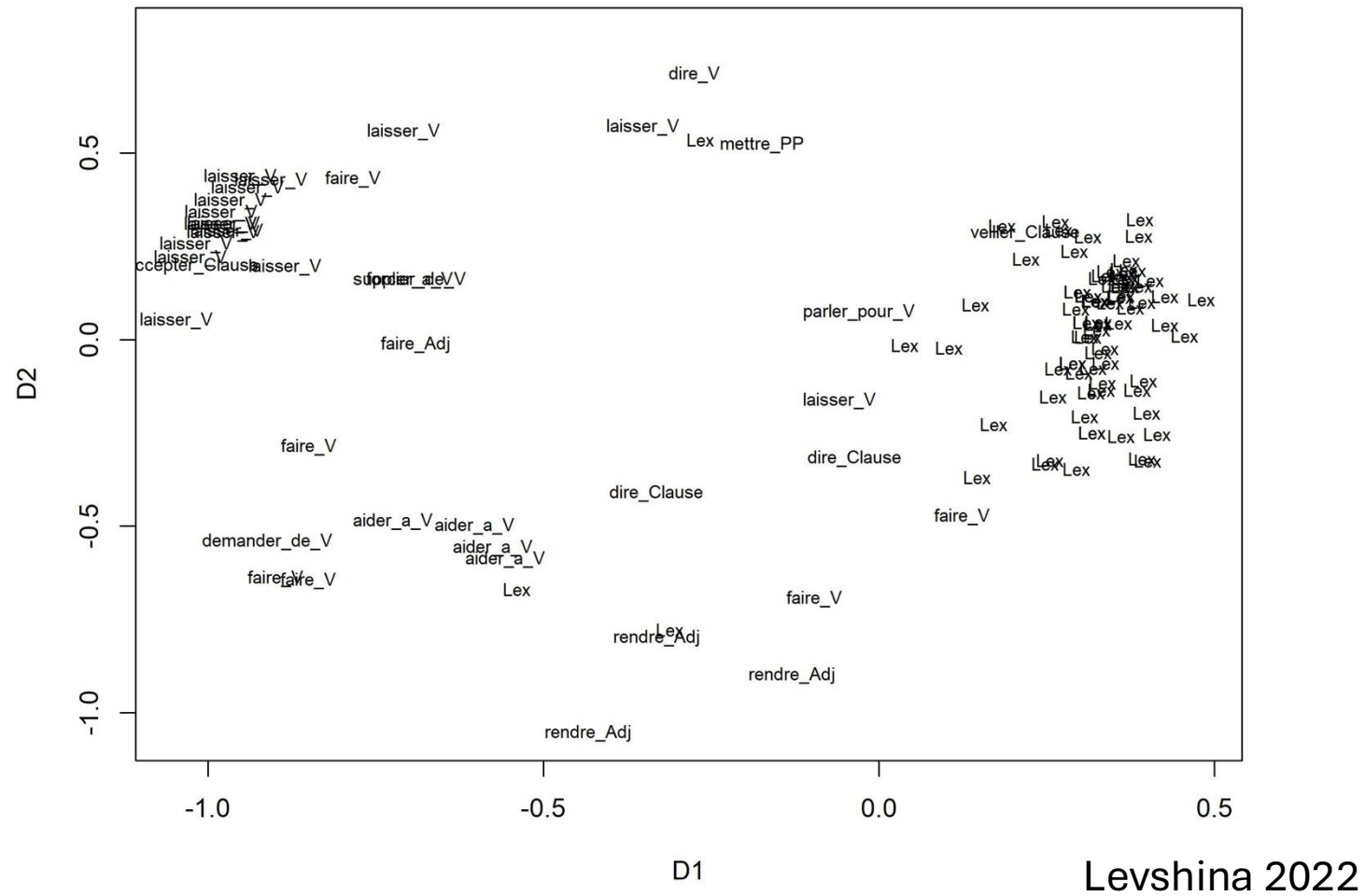
Dimensions 1 & 2: contexts



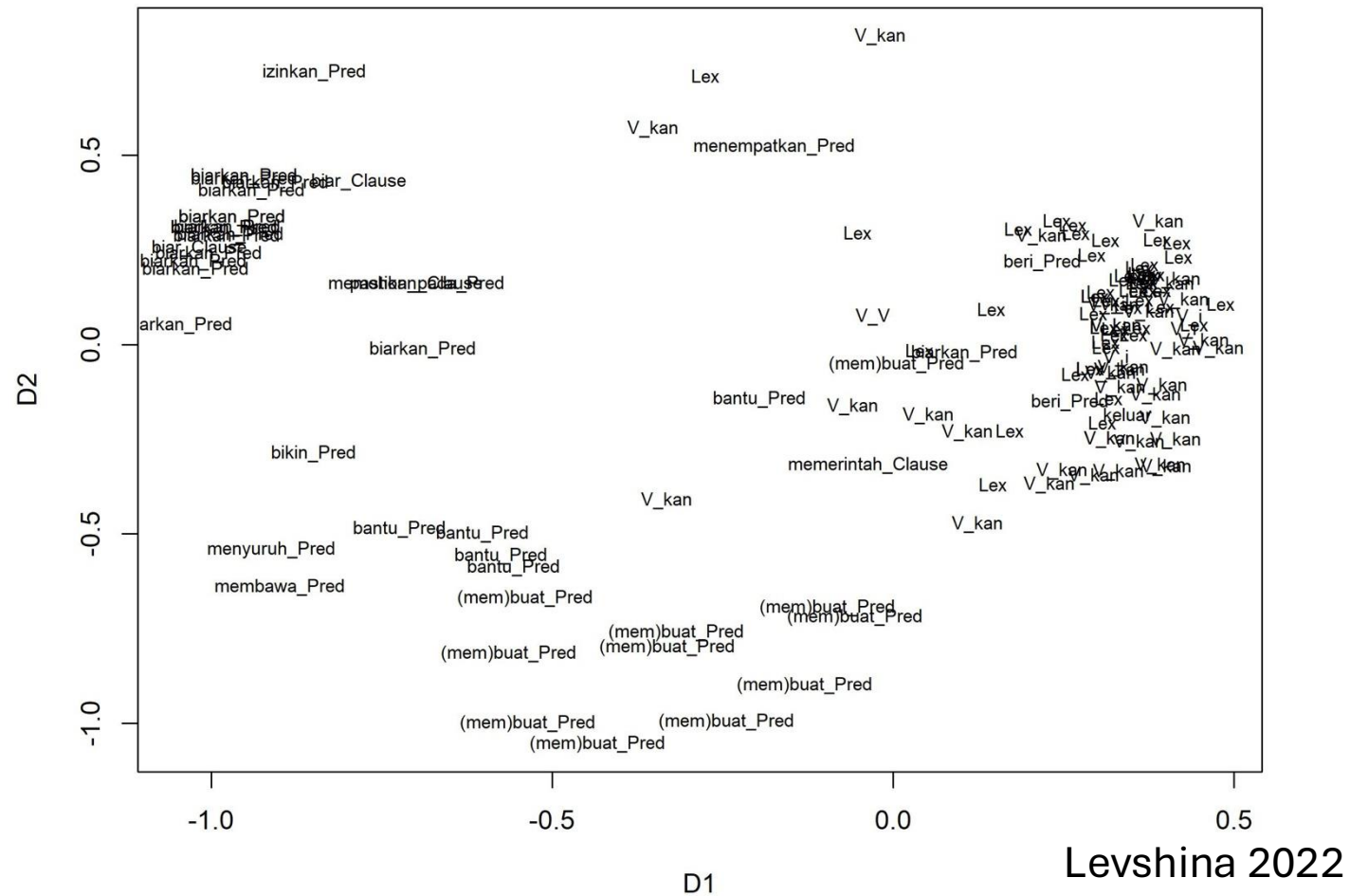
Specific constructions: English



Specific constructions: French



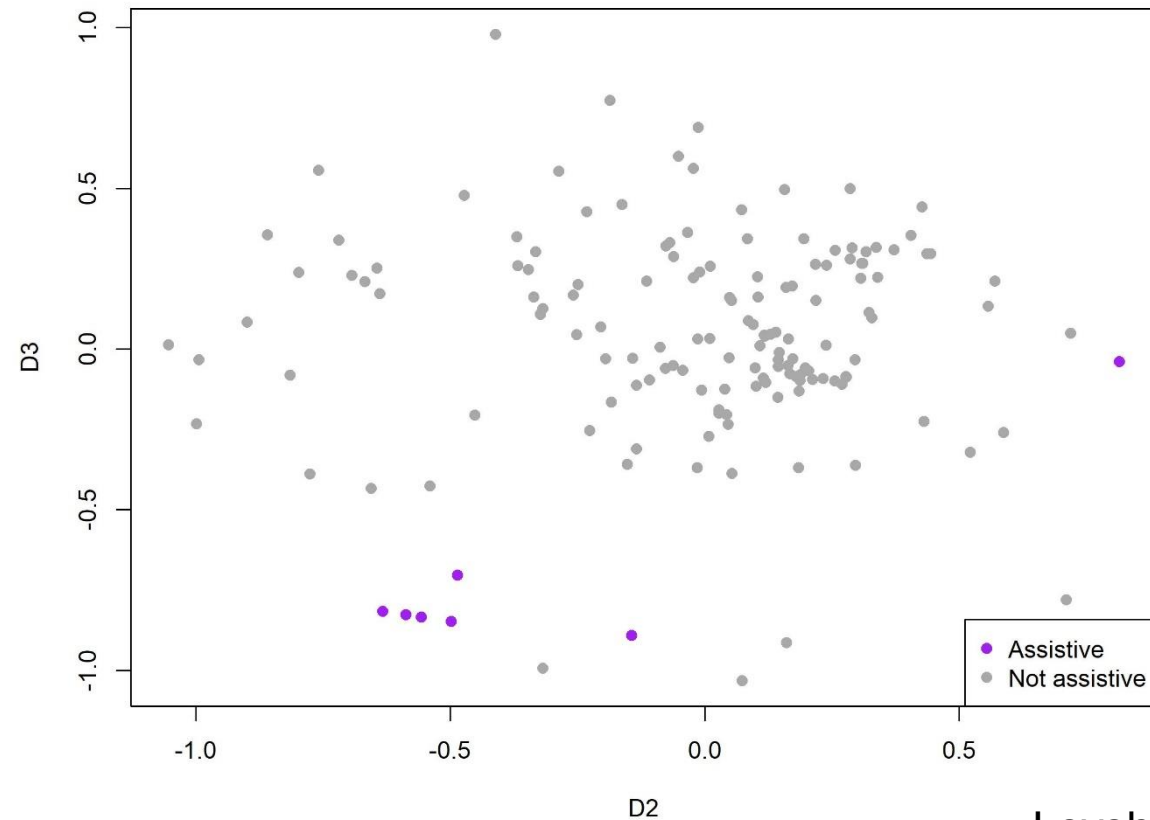
Specific constructions: Indonesian



Interpretation of dimensions (cont.)

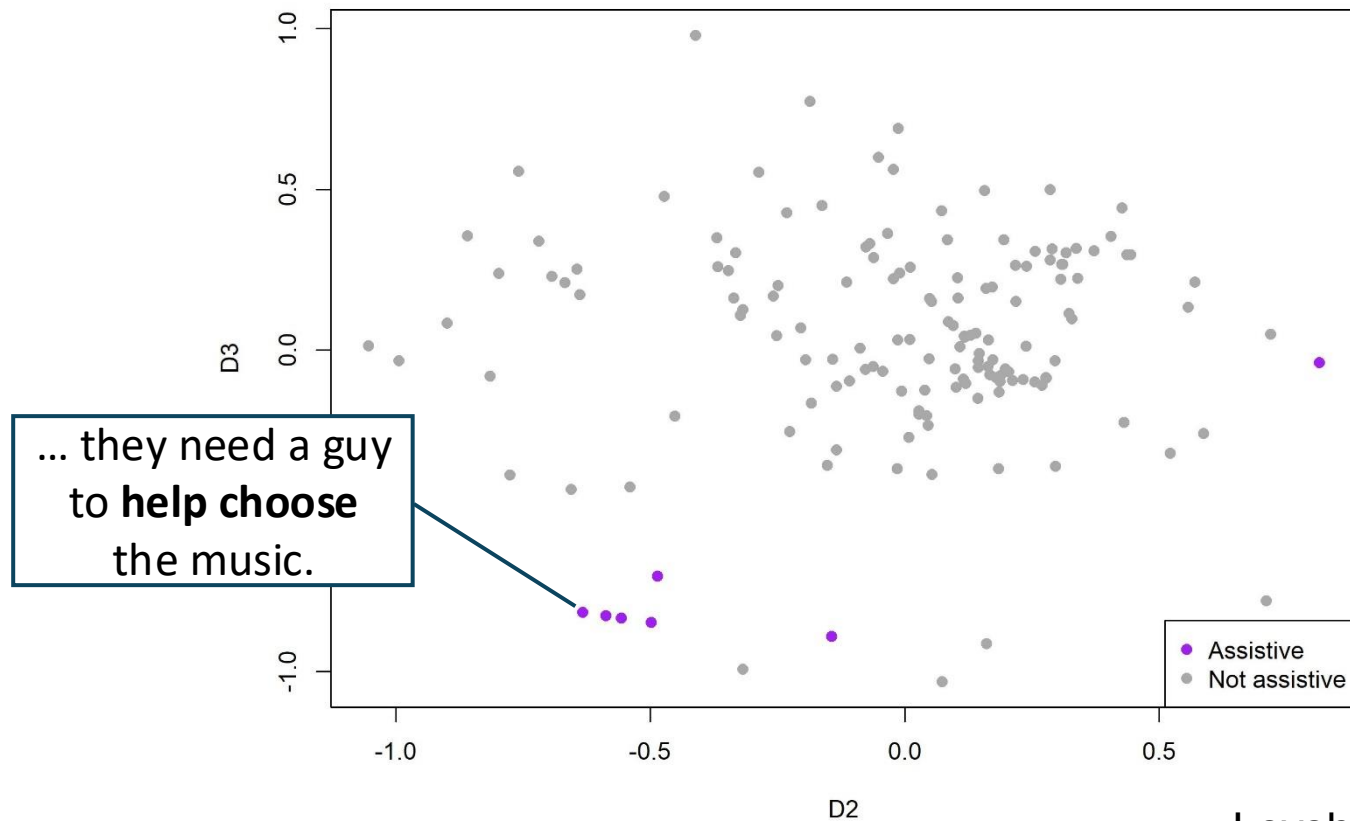
- The R^2 show that the 1st and 2nd dimensions both are the most strongly associated with the contrast between factitive causation (i.e. making X do Y) and permissive causation (i.e. letting X do Y).
- The third dimension is associated with assistive (helping) vs. non-assistive causation.

Dimension 3: assistive vs. non-assistive causation



Levshina 2022

Dimension 3: example



Levshina 2022

The role of data

- Dimensions strongly depend on the source of data (register).
- Films contain many impersonal uses of causatives (Wiezbicka 2006)
 - let of shared information (let me know)
 - let of cooperative dialogue (let me make this announcement)
 - let of cooperative interaction (let me make a cup of tea for you)
 - etc.
- We don't have such predominance of letting and helping when we look at semantic distinctions found in grammars.
- Corpus-based typology deals with doculects!

Steps

1. Collect the data (fictitious example)

	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

Comparative concepts (cf. Haspelmath 2010)

2. Compute the proportion of dissimilar values between the rows (sutations). They will serve as distances.

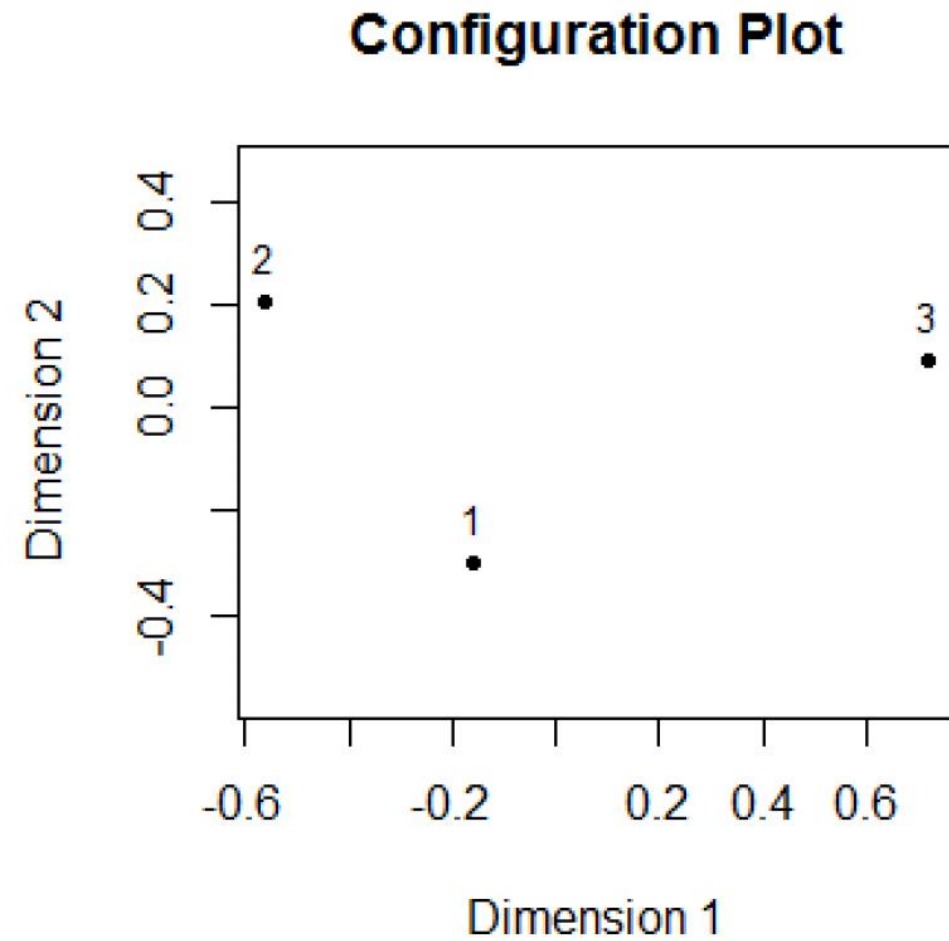
	Lang1	Lang2	Lang3	Lang4	Lang5
Sit1	bla	qu	da	nina	haha
Sit2	bla	qu	da	nana	hihi
Sit3	bla	qa	ta	nina	hehe

$$D(1,2) = 2/5 = 0.4$$

$$D(1,3) = 3/5 = 0.6$$

$$D(2,3) = 4/5 = 0.8$$

3. Perform a MDS analysis.



Intepretation

- The closer two points on the map, the more overlapping constructions they share across the languages.
- Following the isomorphism principle (same function → same form), the corresponding functions/meanings/situations are more semantically similar if more authors of the doculectschose identical constructions to represent these functions/meaning/situations.

Diagnostics

- The less stress (unexplained variance) the better.
- Use a scree plot (stress value) for choosing the number of dimensions.

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Languages

- Indo-European
 - Germanic: Dutch, English, German, Norwegian, Swedish
 - Romance: French, Italian, Portuguese, Romanian, Spanish
 - Slavic: Bulgarian, Czech, Polish, Russian, Slovene
- Uralic:
 - Finnic: Estonian, Finnish
 - Ugric: Hungarian

ParTy corpus: Films for this case study



Examples of analytic causatives

- English: *make*+ Vinf, *let*+ Vinf, *have*+ Vinf, *cause*+ *to*-Vinf
- German: *lassen*+ Vinf
- Dutch: *laten*+ Vinf, *doen*+ Vinf
- Russian: *zastavljat'* “force” + Vinf, *davat'* “give” + Vinf
- French: *faire*+ Vinf, *laisser*+ Vinf
- Romanian: *face*+ *să*+ Vsubj, *lasă*+ *să*+ Vsubj

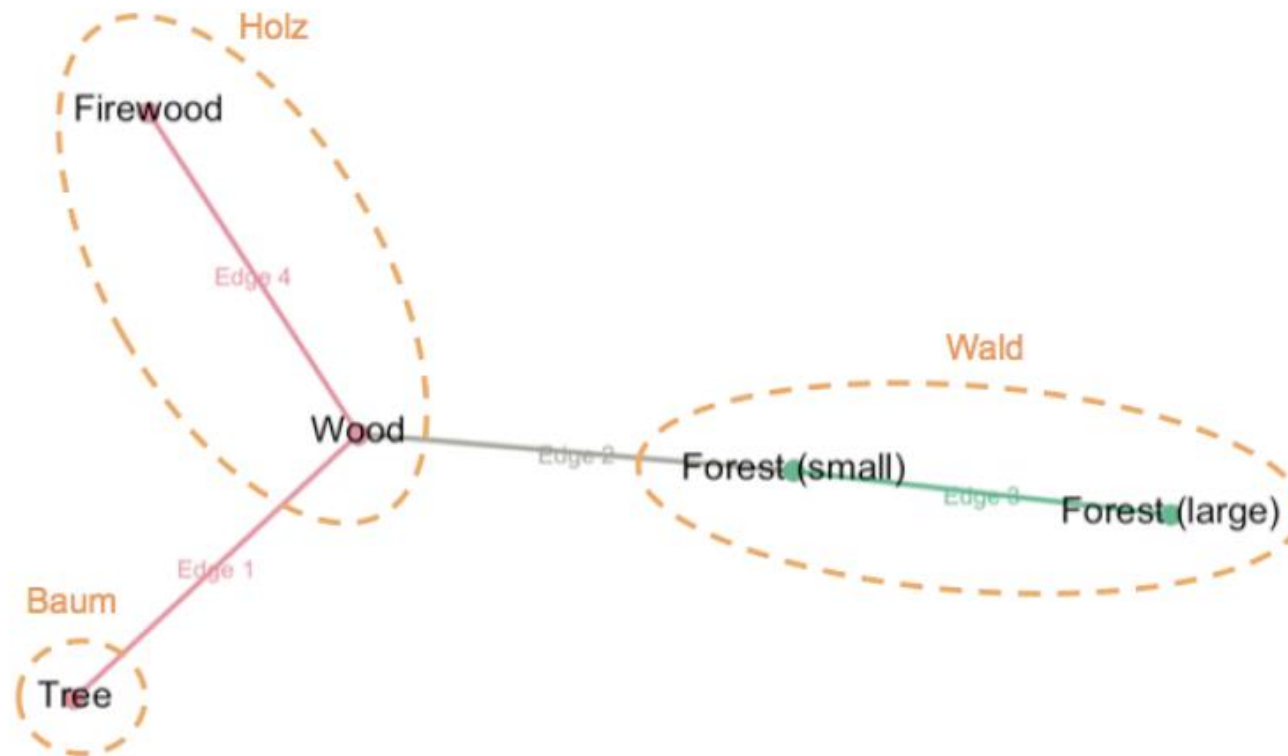
Examples

- Context A

- ENG: And we make them do it... ...or we kill them. **make**
- ITA: E glielo facciamo fare ... o lo uccidiamo. **fare**
- CZE: Donutíme je to udělat, nebo je zabijeme. **donutit**

- Context B

- ENG: Pick up someone my height and build and make them believe it is me. **make**
- ITA: Individua una della mia corporatura e fa credere loro che sia io. **fare**
- CZE: Vyber někoho, kdo je mi podobn. a přesvědč je, že jsem to já. **NA**



from Georgakopoulos & Polis 2018, based on Hjelmslev 1961